Xiaohua Cai

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Software Engineer with nearly 3 years of professional software development experience building full-stack web applications, backend services, REST APIs, and data-driven systems. Additional experience developing Python-based data analysis and automation tools. Skilled in React, TypeScript, JavaScript, Python, PHP, SQL, and relational databases, with experience in cloud deployments, automated testing, and scalable application architecture. M.S. in Computer Science from Georgia Tech (GPA 4.0/4.0).

EDUCATION

Master of Science in Computer Science, Georgia Institute of Technology, GPA 4.0/4.0 2026
Master of Science in Biotechnology, University of Alberta, GPA 3.7/4.0 2018

SKILLS

Languages: JavaScript, TypeScript, Python, Java, PHP, SQL, HTML/CSS/SASS
Frameworks & Libraries: React, Next.js, Node.js, Express, Laravel, Socket.IO, Material UI, Tailwind CSS, Bootstrap
Databases & APIs: PostgreSQL, MySQL, MongoDB, GraphQL, REST
Infrastructure & Tools: AWS, Docker, Docker Compose, Nginx, Git/GitHub, GitHub Actions, CI/CD, Postman, Ubuntu Linux

WORK EXPERIENCE

Software Development Engineer | Affinity Bridge Inc. December 2022 - June 2024

- Built the Environmental Conventions Index application using Next.js, Typescript and React, developing interactive geographic maps with React Simple Maps and data visualizations with ECharts; Built reusable UI components and custom hooks to reduce code duplication and improve maintainability.
- Built an internal authenticated architecture-review application using Next.js, TypeScript, Postgresql, and NextAuth, replacing Word-based processes with a centralized, authenticated workflow for 20+ users.
- Designed and implemented REST APIs, middleware, relational data models, and caching strategies, reducing redundant data fetching and improving backend performance.
- Developed React frontend features and PHP REST APIs for the NovaSafe application, supporting thousands of users; Optimized SQL queries by restructuring JOINs, refining filtering conditions, and adding targeted indexes, reducing query execution time by approximately 20%, and implemented pagination to improve large-dataset performance.
- Developed 200+ unit and integration tests for React components and end-to-end tests with Playwright, reducing pre-release manual testing time by approximately 25%.
- Supported production deployments and application releases by investigating build, configuration, API, and runtime failures and implementing fixes to improve application stability and release reliability.
- Integrated AI-assisted workflows across code exploration, debugging, testing, and documentation, accelerating development cycles while maintaining engineering quality through code reviews, automated testing, and security validation.

Software Development Engineer | AMI RockChain Inc. January 2022 - December 2022

- Developed backend services using PHP Laravel, including REST APIs, PostgreSQL database schemas, migrations, and CRUD operations following MVC architecture; implemented authentication and role-based access control to secure user data and enforce permissions for across different user roles.
- Built a Python data-enrichment pipeline that integrated 3,000+ CSV and JSON records, matched records across sources, and populated missing customer information, reducing an estimated multi-week manual process to minutes.
- Developed reusable, responsive UI components and user workflows using React and TypeScript, reducing code duplication and improving user experience; built interactive maps and data visualizations using Leaflet, Google Maps API, and ECharts, and integrated the Mailchimp API for subscriber management.

Software Development Engineer (Contract) | Explorer Hop September 2021 - February 2022

- Built a full-stack internal web application using React, TypeScript, Material UI, Node.js, Express, and PostgreSQL; implemented REST APIs, backend business logic, and relational data workflows, and contributed to deploying the application to AWS using Amazon EC2.
- Developed Shopify admin app features using React, Shopify App Bridge, and Polaris; integrated the Shopify GraphQL API and implemented webhooks for product-order events.

Research and Development Scientist | SBI BioEnergy Inc. January 2019 - August 2021

- Developed Python scripts to process, analyze, and visualize experimental data from 10+ lab-scale reactors, improving data analysis efficiency and reducing repetitive manual work.

PROJECTS

Distributed Real-Time Chat Application| React, Node.js, Redis, PostgreSQL, Nginx, Docker 2026

- Re-architected a previously built single-server real-time chat application into a horizontally scalable distributed system, running multiple Socket.IO backend instances behind an Nginx load balancer.
- Implemented Redis Pub/Sub for cross-instance event propagation and Redis-backed shared room/presence state; persisted message history in PostgreSQL and containerized the distributed stack with Docker Compose.
- Built a concurrent WebSocket load-testing script and validated 200 simultaneous clients and 40,000 message deliveries with zero errors; verified continued availability for new connections after terminating one backend instance.